Time steps of the Hamiltonian and viscous steps for $h/a \gg 1$

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Abstract

The Hamiltonian step resolves the fields on the smoothing length h; the viscous (stochastic) step acts on the particles, spacing a. We list the rates that limit each step, work out how they scale for $h/a \gg 1$, and propose a splitting with one long Hamiltonian step and several viscous sub-steps at fixed positions. The main finding: in the present scheme the viscous rates are set by the bond range h_ψ, which is h, not a; the spacing a enters through the thermal velocity of a particle, which sets the stability limit. Whether many viscous sub-steps are needed depends on choosing $h_\psi \sim a$. For discussion; nothing is implemented.

I. THE RATES

Symbols: sound speed c_s, kinematic viscosity $\nu = \eta/w$ (and $\kappa_b = 2\eta + \zeta_{\text{eff}}$, the largest viscosity carried by the bonds), smoothing length h, particle spacing a, bond range $h_\psi \le h$, particle volume $V_a \simeq \frac{\sqrt{3}}{2}a^2$ (hexagonal lattice), entropy per particle $s_a = SV_a$. Numbers are for the present runs: $h = 0.05$, $h/a = 2$, $h_\psi = h$, $\eta/s = 0.2$, $S = 10^6$.

a. Hamiltonian step. The forces and velocities are smoothed over h, so the fastest resolved motion is a sound wave with $k \sim 1/h$,

$$\Omega_H = \frac{c_s}{h} = 11.5. \tag{1}$$

Particle-scale momentum modes ($kh \gg 1$) enter the smoothed fields with the weight $\tilde{W}(kh) \ll 1$: in the Hamiltonian step they are carried along, not oscillating. The leapfrog needs $\Omega_H \Delta t \lesssim 1$; the code uses $\Delta t = \frac{1}{2} \times 0.2\, h/c_s = 8.66 \times 10^{-3}$, i.e. $\Omega_H \Delta t = 0.10$.

b. Viscous step. The bond form damps a mode of wavenumber k at

$$\Gamma(k) = \nu k^2 \quad (kh_\psi \ll 1), \qquad \Gamma(k) \to \Gamma_{\max} \simeq 2\,\frac{\kappa_b}{w}\,\frac{c_d}{h_\psi^2} \quad (kh_\psi \gtrsim 1), \tag{2}$$

the saturation coming from $\sum_b w_{ab} \le c_d/h_\psi^2$, $c_2 = 7/2$ (`changerep.tex`: the Gershgorin bound is $4\kappa_b c_d/(w h_\psi^2) = 29$, the measured zone-edge rate about half of it). The resolved modes ($kh < 1$) relax at $\nu k^2 \le \nu/h^2 = 0.88$, the unresolved ones at up to $\Gamma_{\max} \simeq 14$.

c. Acceptance. For a residual move the entropy change of the star is, to leading order, $\Delta S = \sum_b \Delta\beta_{ab} \cdot \boldsymbol{y}_{ab}$ plus a quadratic cost, with $\langle y^2 \rangle = 2\Delta t\, c_{ab}$ and the thermal $\Delta\beta_{ab}$ of two particles. Summing over the bonds,

$$\langle \Delta S^2 \rangle \simeq 8\,\frac{\kappa_b}{w}\,\Delta t \sum_b w_{ab} \sim \Gamma_{\max}\Delta t, \qquad 1 - \text{acceptance} \simeq c\,\sqrt{\Gamma_{\max}\Delta t}, \tag{3}$$

independent of a at fixed h_ψ. Measured: residual acceptance 0.88 at $\Gamma_{\max}\Delta t \simeq 0.06$ (split half steps) and 0.77 at 0.12 (merged), so $c \simeq 0.5$–0.65 (the rejection grows somewhat faster than the square root). The Metropolis sweep samples $e^{\mathbb{S}}$ for *any* Δt: a long step is not unstable, it is rejected, and the rejected moves damp nothing. So the viscous step needs $\Gamma_{\max}\Delta t \lesssim 0.1$ (acceptance $\gtrsim 0.8$) for its rates, not for its stability.

d. The particle scale. The spacing enters through the thermal velocity of one particle,

$$\delta v^2 = \frac{T}{V_a\, w} \propto \frac{1}{a^2} \qquad (\text{fixed } S, T), \tag{4}$$

$\delta v = 0.043$ at $S = 10^6$, $h/a = 2$. The stability scan (`nersc/LOG.md`) puts the $\lambda = 1$ melting threshold between $\delta v = 0.078$ and 0.14 at $h/a = 2$; at $h/a = 3$ and $S = 10^5$ ($\delta v = 0.20$, the same smoothed fluctuation $\delta v\, a/h$ as $h/a = 2$) the lattice melted earlier, $t \approx 11$ against 28. So the particle velocity, not the smoothed one, appears to control melting.

II. THE LIMIT $h/a \gg 1$

Hold h, c_s, ν and T fixed and let $a \to 0$. The number of neighbours within $2h$ grows as $N_{\rm nb} \simeq \frac{8\pi}{\sqrt{3}}(h/a)^2$.

a. Bond range $h_\psi = h$ (the present scheme). Nothing in eq. (2) or eq. (3) depends on a:

$$\frac{\Gamma_{\max}}{\Omega_H} \simeq 2c_d\,\frac{\kappa_b}{\eta}\,\frac{1}{\mathrm{Re}_h} = 1.25, \qquad \mathrm{Re}_h = \frac{c_s h}{\nu} = 13, \tag{5}$$

a fixed ratio. The two steps then want comparable time steps for all h/a, and both cost $\propto N\, N_{\rm nb} \propto (h/a)^4$ per area. There is no gain from sub-stepping the viscous step. At the present numbers the Hamiltonian step is run far below its stability limit ($\Omega_H \Delta t = 0.10$), while the viscous step is near its target ($\Gamma_{\max}\Delta t = 0.12$ merged); the viscous step is also the more expensive one. The limit is instead set by eq. (4): at fixed S the particle velocity grows as h/a, from 0.043 at $h/a = 2$ ($S = 10^6$) into the range 0.08–0.14 where the $h/a = 2$ runs melt by $h/a \simeq 4$–6, unless S (i.e. s_a) grows with $(h/a)^2$.

b. Bond range $h_\psi = b\,a$ (local bonds, $b = O(1)$). Now

$$\Gamma_{\max} \simeq 2\,\frac{\kappa_b}{w}\,\frac{c_d}{b^2 a^2}, \qquad \frac{\Gamma_{\max}}{\Omega_H} \simeq \frac{2c_d}{b^2}\,\frac{\kappa_b}{\eta}\,\frac{1}{\mathrm{Re}_h}\left(\frac{h}{a}\right)^2 \gg 1, \tag{6}$$

the regime of a long Hamiltonian step and many viscous steps. The modes separate:

- the resolved modes, $k < 1/h$, relax at $\nu k^2 \le \nu/h^2 \ll \Omega_H$; the bond form gives the right ν for smooth fields for any h_ψ (its normalization $\sum_b w_{ab}\boldsymbol{r}_{ab}\boldsymbol{r}_{ab}^T = \mathbb{1}$);
- the unresolved modes, $1/h < k < 1/a$, $(h/a)^2$ of them per resolved mode, relax at up to $\Gamma_{\max} \propto 1/a^2$, much faster than anything the Hamiltonian step does.

In the limit the unresolved modes are a fast heat bath in equilibrium at the current positions; the resolved fields obey fluctuating hydrodynamics on the scale h, their noise the sum of $(h/a)^2$ independent particle noises. The Hamiltonian step needs $\Omega_H \Delta t_H \lesssim 0.2$; the viscous step needs $\Gamma_{\max}\Delta t_V \lesssim 0.1$ by eq. (3); the number of viscous sub-steps is

$$n_V = \frac{\Delta t_H}{\Delta t_V} \simeq 2\,\frac{\Gamma_{\max}}{\Omega_H} \propto \left(\frac{h}{a}\right)^2. \tag{7}$$

A star then has $O(b^2)$ bonds instead of $O((h/a)^2)$, so a sub-step costs $\propto N$, and n_V sub-steps cost $\propto N(h/a)^2$, the same order as one Hamiltonian step.

III. PROPOSAL

a. The splitting. One step of length Δt_H,

$$\left[S\left(\tfrac{\Delta t_H}{2n_V}\right)\right]^{n_V} H(\Delta t_H) \left[S\left(\tfrac{\Delta t_H}{2n_V}\right)\right]^{n_V}, \tag{8}$$

with the two blocks of consecutive steps merged as now. During the viscous sub-steps the positions do not move, so everything that depends only on them is computed once per Hamiltonian step: the volumes V_a, the particle energies e_a, the stars (comoving weights, coefficients) and the kernel stars of the gradient correction. A sub-step is then only the two sweeps and the back map, about half of the present stochastic step. (Strictly the stars depend on the particle velocities through the comoving metric; for $v \ll 1$ freezing them over Δt_H is an error of order Δv.)

b. Choices.

1. $\Delta t_H = C\,h/c_s$, with the Courant number C raised from the present 0.1 until the resolved sound and shear rates or the stability degrade (to be measured).

2. $n_V = \lceil \Gamma_{\max} \Delta t_H / 0.1 \rceil$, i.e. a residual acceptance $\gtrsim 0.8$ per sub-step, with $\Gamma_{\max}$ half the bound of `compute_dt`.

3. h_ψ: keep h (then $n_V = 1$–2, and the gain is only the larger C), or go to $h_\psi \simeq a$–$2a$ (local bonds, cheaper stars, $n_V \propto (h/a)^2$). The second changes the dissipative function at $kh_\psi \gtrsim 1$, so the deterministic tests of `changerep.tex` and the frozen-lattice tests of `stochrep.tex` would have to be redone.

c. Tests. On a frozen lattice nothing changes: sub-steps are exact Metropolis steps of $e^{\mathbb{S}}$. With the Hamiltonian step on: (i) the sound and shear decay rates of resolved modes against Δt_H and n_V (splitting error $O(\Delta t_H^2)$); (ii) equilibrium statics, the energy drift and the melting threshold against C; (iii) cost per unit time.

d. Expected gain. At $h/a = 2$, $h_\psi = h$: from C alone, perhaps a factor 2–3 if $C \simeq 0.3$ is stable. For $h/a \gg 1$ with local bonds: the cost per unit time becomes $\propto N(h/a)^2$ instead of $\propto N(h/a)^4$ with $h_\psi = h$.

IV. POINTS FOR DISCUSSION

- In the present scheme ($h_\psi = h$) the viscous step is not limited by a; is the intended scheme one with local bonds, $h_\psi \sim a$?

- At fixed S the particle velocity grows as h/a (eq. (4)), into the melting range by $h/a \simeq 4$–6. $h/a \gg 1$ needs either $s_a \propto (h/a)^{-2} S$ kept above a few hundred (larger S) or a scheme whose stability does not depend on δv; the stability scan at fixed δv and varying h/a would decide which velocity matters.

- With $\Gamma_{\max}\Delta t_V \ll 1$ the unresolved modes are in equilibrium at every Hamiltonian step. Instead of n_V sub-steps one could redraw them directly from $e^{\mathbb{S}}$ at fixed resolved fields (a heat bath for the residual part), and use sub-steps only for the resolved part, which is slow.